

Disproof of Conjecture 000000000499

October 2, 2026

Abstract

Verdict: FALSE. The conjecture asserts that the vicious-walker hitting probability p_n equals, at the corresponding endpoints, the joint density of the ordered eigenvalues of an $n \times n$ GUE matrix times the explicit Vandermonde factor $\prod_{i < j} (y_j - y_i - (x_j - x_i))$, and that this identity holds for all times $T > 0$. The right-hand side contains no T and is therefore a single constant, while at the fixed endpoints $(x_1, x_2) = (0, 4)$, $(y_1, y_2) = (1, 3)$ for $n = 2$ an exact brute-force enumeration gives the distinct values $p(1) = 1/4$, $p(2) = 0$ and $p(3) = 1/8$. A constant cannot equal three different numbers, so the identity cannot hold for all $T > 0$.

1 The conjecture

Let p_n denote the probability that n nonintersecting simple random walks go from fixed starting points $x = (x_1, \dots, x_n)$ to fixed endpoints $y = (y_1, \dots, y_n)$ without ever intersecting. The conjecture states that

$$p_n = \rho_{n,\text{ord}}^{\text{GUE}}(y) \cdot \prod_{i < j} (y_j - y_i - (x_j - x_i)),$$

where $\rho_{n,\text{ord}}^{\text{GUE}}$ is the joint density of the ordered eigenvalues of an $n \times n$ GUE matrix, and that this holds *for all times* $T > 0$.

2 Attack: same endpoints, different times

Take $n = 2$ with starts $x = (0, 4)$ and endpoints $y = (1, 3)$; each walk moves ± 1 per step, and all 4^T joint step sequences are equally likely. We enumerate every joint step sequence exactly and count those for which both walks end at $(1, 3)$ and the strict order $w_1(t) < w_2(t)$ holds at every time $t = 0, \dots, T$.

T	good / total	$p(T)$
1	1/4	1/4
2	0/16	0
3	8/64	1/8
4	0/256	0
5	75/1024	75/1024

- $T = 1$: the only surviving joint path is $(+1, -1)$, taking $(0, 4)$ to $(1, 3)$ in one step; $p(1) = \frac{1}{4}$.
- $T = 2$: parity. A displacement of $+1$ in two ± 1 steps is impossible, so no joint path reaches the endpoints; $p(2) = 0$.
- $T = 3$: exactly 8 of the 64 joint paths survive; for instance $w_1: 0 \rightarrow 1 \rightarrow 0 \rightarrow 1$ with $w_2: 4 \rightarrow 3 \rightarrow 4 \rightarrow 3$. Hence $p(3) = \frac{8}{64} = \frac{1}{8}$.

3 Why the conjecture fails

The proposed right-hand side evaluated at these endpoints is

$$\rho_{2,\text{ord}}^{\text{GUE}}(1, 3) \cdot \left((3 - 1) - (4 - 0) \right) = -2 \rho_{2,\text{ord}}^{\text{GUE}}(1, 3),$$

which *does not depend on T* . It is a single fixed constant c ; moreover $c \neq 0$, since the ordered GUE density at two distinct points is strictly positive and the Vandermonde-type factor equals $\pm 2 \neq 0$ (under the literal reading c is even negative, while a probability satisfies $p_n \geq 0$).

The left-hand side, however, takes the distinct values $\frac{1}{4}, 0, \frac{1}{8}$ at the same endpoints as T varies. A single constant cannot equal three different numbers; in particular $p(1) = \frac{1}{4} \neq \frac{1}{8} = p(3)$, while the right-hand side is the same number at $T = 1$ and $T = 3$. Therefore the identity fails for some $T > 0$, and the conjecture is **false**.

4 Verification

- `reproduce.py`: standalone exact recomputation (standard library only), recomputing the table above with rational arithmetic.
- `lean4/`: a core-Lean (no Mathlib) development that explicitly enumerates the 4, 16 and 64 joint step sequences for $T = 1, 2, 3$ and proves, with zero axioms and no `sorry`, the counts $C_1 = 1$, $C_2 = 0$, $C_3 = 8$ and the cross-multiplied inequality $C_1 \cdot 64 \neq C_3 \cdot 4$ (i.e. $\frac{1}{4} \neq \frac{1}{8}$). An axiom audit (`Check.lean`) confirms each theorem “does not depend on any axioms”.